Tutorial - 2_3

Que 1 :	Student Record Management (Class, Object, Encapsulation)
Code :	

```
class Student {
  private int rollNo;
  private String name;
  private double marks;

  public void setRollNo(int rollNo) {
    this.rollNo = rollNo;
  }

  public int getRollNo() {
    return rollNo;
  }

  public void setName(String name) {
    this.name = name;
  }

  public String getName() {
    return name;
  }

  public void setMarks(double marks) {
    if (marks >= 0 && marks <= 100) {
      this.marks = marks;
    } else {
      System.out.println("Invalid marks. Marks must be between 0 and 100.");
    }
  }

  public double getMarks() {
    return marks;
  }

  public void displayDetails() {
    System.out.println("Roll No: " + rollNo);
    System.out.println("Name: " + name);
    System.out.println("Marks: " + marks);

    if (marks >= 90)
      System.out.println("Grade: A");
    else if (marks >= 75)
      System.out.println("Grade: B");
    else if (marks >= 60)
      System.out.println("Grade: C");
    else if (marks >= 40)
      System.out.println("Grade: D");
    else
      System.out.println("Grade: F");

    System.out.println();
  }
```

```
}

public class Program1 {
  public static void main(String[] args) {

    Student s1 = new Student();
    s1.setRollNo(1);
    s1.setName("Radha");
    s1.setMarks(85);

    Student s2 = new Student();
    s2.setRollNo(2);
    s2.setName("Ravi");
    s2.setMarks(92);

    Student s3 = new Student();
    s3.setRollNo(3);
    s3.setName("Amit");
    s3.setMarks(68);

    s1.displayDetails();
    s2.displayDetails();
    s3.displayDetails();
  }
}
```

Output :

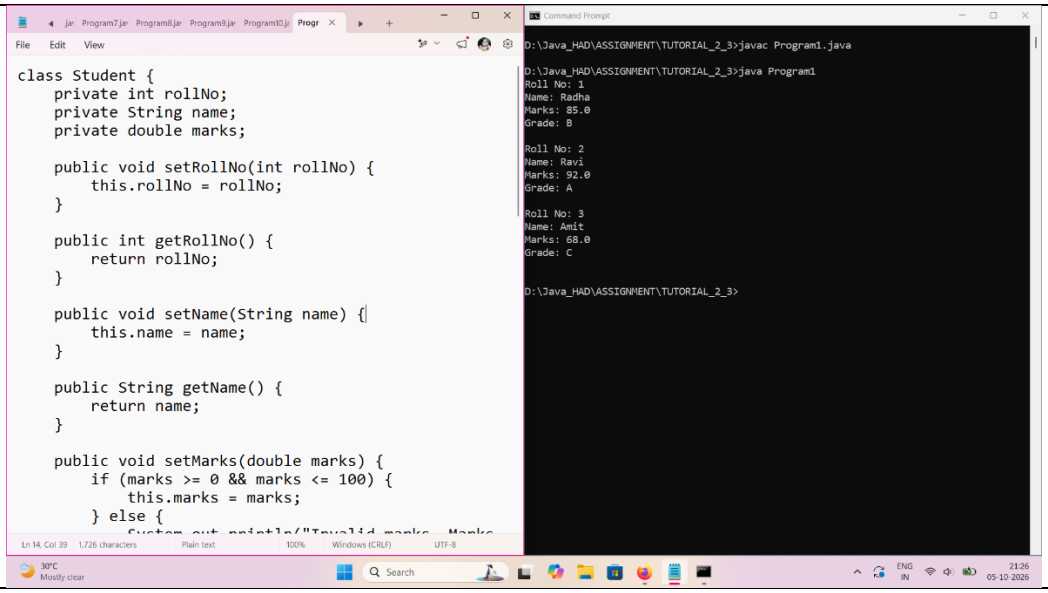
```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program1.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program1
Roll No: 1
Name: Radha
Marks: 85.0
Grade: B

Roll No: 2
Name: Ravi
Marks: 92.0
Grade: A

Roll No: 3
Name: Amit
Marks: 68.0
Grade: C
```

Main Output :



```
class Student {
    private int rollNo;
    private String name;
    private double marks;

    public void setRollNo(int rollNo) {
        this.rollNo = rollNo;
    }

    public int getRollNo() {
        return rollNo;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getName() {
        return name;
    }

    public void setMarks(double marks) {
        if (marks >= 0 && marks <= 100) {
            this.marks = marks;
        } else {
```

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program1.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program1
Roll No: 1
Name: Radha
Marks: 85.0
Grade: B

Roll No: 2
Name: Ravi
Marks: 92.0
Grade: A

Roll No: 3
Name: Amit
Marks: 68.0
Grade: C

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>
```

Que 2 : **Bank Account System (Encapsulation, Validation)**

Code :

```
class BankAccount {
  private double balance;

  public void deposit(double amount) {
    if (amount > 0) {
      balance += amount;
      System.out.println("Deposit successful: " + amount);
    } else {
      System.out.println("Deposit failed: Amount cannot be negative or zero.");
    }
  }

  public void withdraw(double amount) {
    if (amount <= 0) {
      System.out.println("Withdrawal failed: Amount must be positive.");
    } else if (amount > balance) {
      System.out.println("Withdrawal failed: Insufficient balance.");
    } else {
      balance -= amount;
      System.out.println("Withdrawal successful: " + amount);
    }
  }

  public void checkBalance() {
    System.out.println("Current Balance: " + balance);
  }
}

public class Program2 {
  public static void main(String[] args) {

    BankAccount account = new BankAccount();

    account.deposit(5000);
    account.checkBalance();

    account.withdraw(2000);
    account.checkBalance();
```

```
        account.deposit(-500);

        account.withdraw(5000);

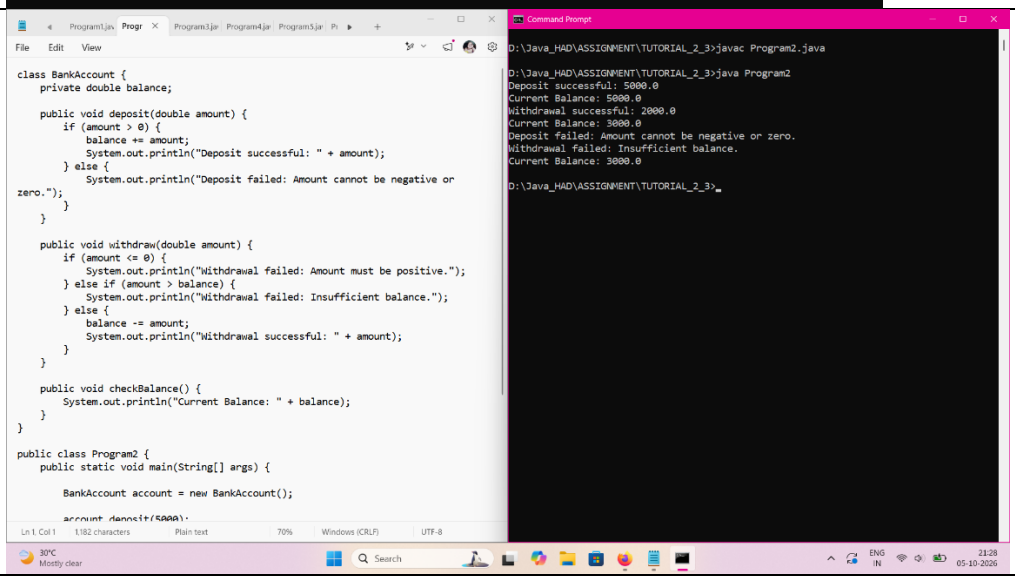
        account.checkBalance();
    }
}
```

Output :

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program2.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program2
Deposit successful: 5000.0
Current Balance: 5000.0
Withdrawal successful: 2000.0
Current Balance: 3000.0
Deposit failed: Amount cannot be negative or zero.
Withdrawal failed: Insufficient balance.
Current Balance: 3000.0
```

Main Output :



Que 3 : **Employee Counter (Static Variable, Static Method)**

Code :

```
class Employee {
    static int count = 0;

    Employee() {
        count++;
    }

    static int getCount() {
        return count;
    }
}

public class Program3 {
    public static void main(String[] args) {

        Employee e1 = new Employee();
        System.out.println("Employees: " + Employee.getCount());

        Employee e2 = new Employee();
```

```
        System.out.println("Employees: " + Employee.getCount());

        Employee e3 = new Employee();
        System.out.println("Employees: " + Employee.getCount());

        Employee e4 = new Employee();
        System.out.println("Employees: " + Employee.getCount());

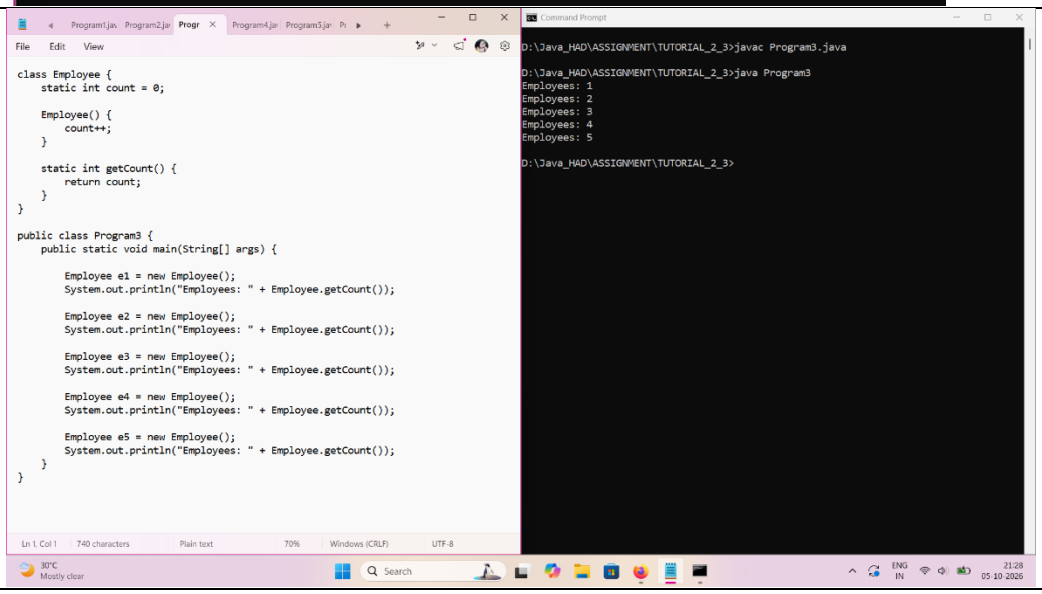
        Employee e5 = new Employee();
        System.out.println("Employees: " + Employee.getCount());
    }
}
```

Output :

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program3.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program3
Employees: 1
Employees: 2
Employees: 3
Employees: 4
Employees: 5
```

Main Output :



Que 4 : **Area Calculator (Method Overloading)**

Code :

```
class Shape {

    double area(double radius) {
        return 3.14 * radius * radius;
    }

    double area(double length, double breadth) {
        return length * breadth;
    }

    double area(double base, double height, boolean isTriangle) {
        return 0.5 * base * height;
    }
}

public class Program4 {
```

```
    public static void main(String[] args) {

        Shape s = new Shape();

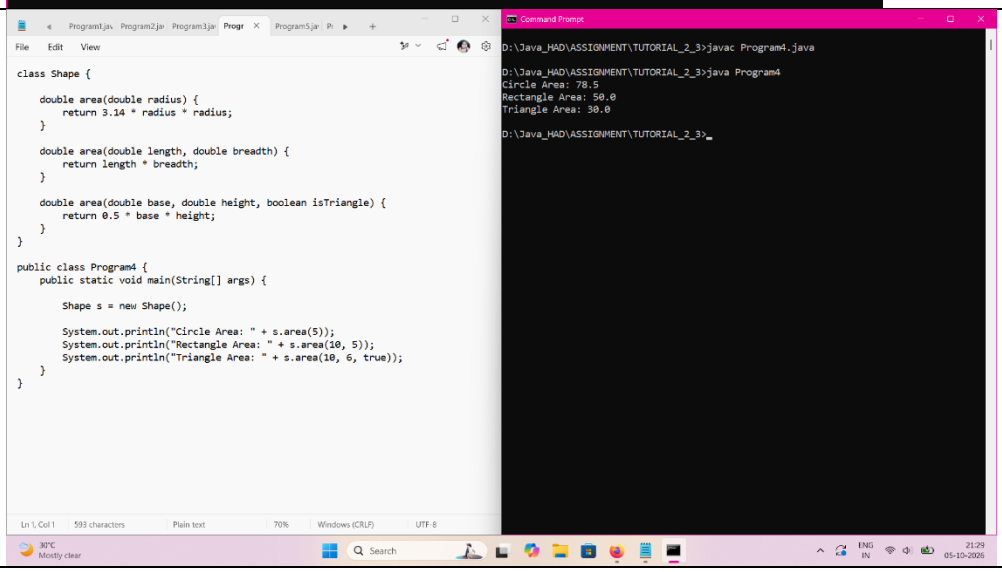
        System.out.println("Circle Area: " + s.area(5));
        System.out.println("Rectangle Area: " + s.area(10, 5));
        System.out.println("Triangle Area: " + s.area(10, 6, true));
    }
}
```

Output :

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program4.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program4
Circle Area: 78.5
Rectangle Area: 50.0
Triangle Area: 30.0
```

Main Output :



Que 5 : Constructor Overloading and Constructor Chaining (this())

Code :

```
class Box {
  double length;
  double breadth;
  double height;

  // Default constructor
  Box() {
    this(1, 1, 1);
  }

  // Parameterized constructor
  Box(double length, double breadth, double height) {
    this.length = length;
    this.breadth = breadth;
    this.height = height;
  }

  // Single argument constructor for cube
  Box(double side) {
    this(side, side, side);
  }

  void displayVolume() {
```

```
        double volume = length * breadth * height;
        System.out.println("Volume: " + volume);
    }
}

public class Program5
{
    public static void main(String[] args) {

        Box b1 = new Box();
        b1.displayVolume();

        Box b2 = new Box(10, 5, 2);
        b2.displayVolume();

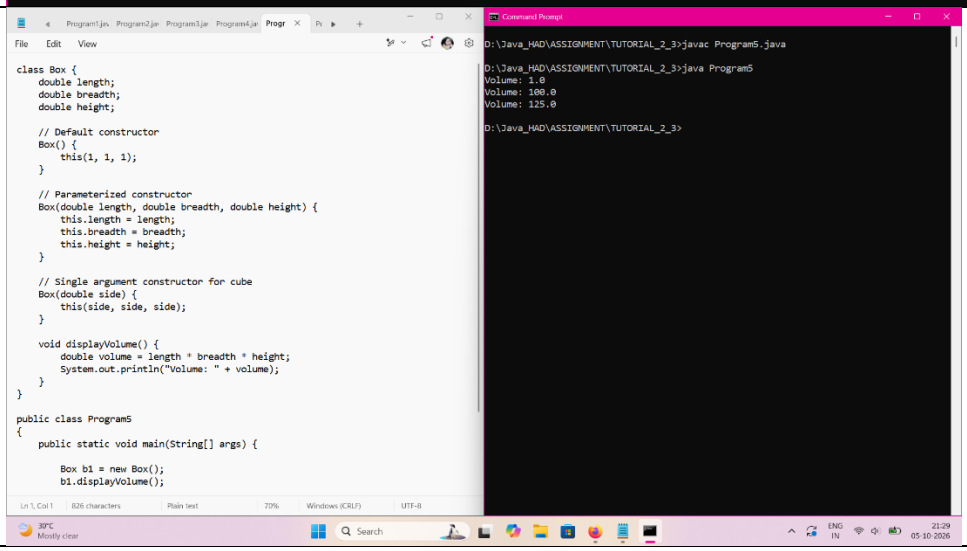
        Box b3 = new Box(5);
        b3.displayVolume();
    }
}
```

Output :

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program5.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program5
Volume: 1.0
Volume: 100.0
Volume: 125.0
```

Main Output :



Que 6 : **Resolving Variable Shadowing (this Keyword)**

Code :

```
class Person {
    String name;
    int age;

    Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    void setDetails(String name, int age) {
        this.name = name;
        this.age = age;
```

```
  }

  void display() {
    System.out.println("Name: " + name);
    System.out.println("Age: " + age);
  }
}

public class Program6
{
  public static void main(String[] args) {

    Person p = new Person("Radha", 25);

    p.display();

    p.setDetails("Krishna", 30);

    System.out.println("\nAfter changing details:");
    p.display();
  }
}
```

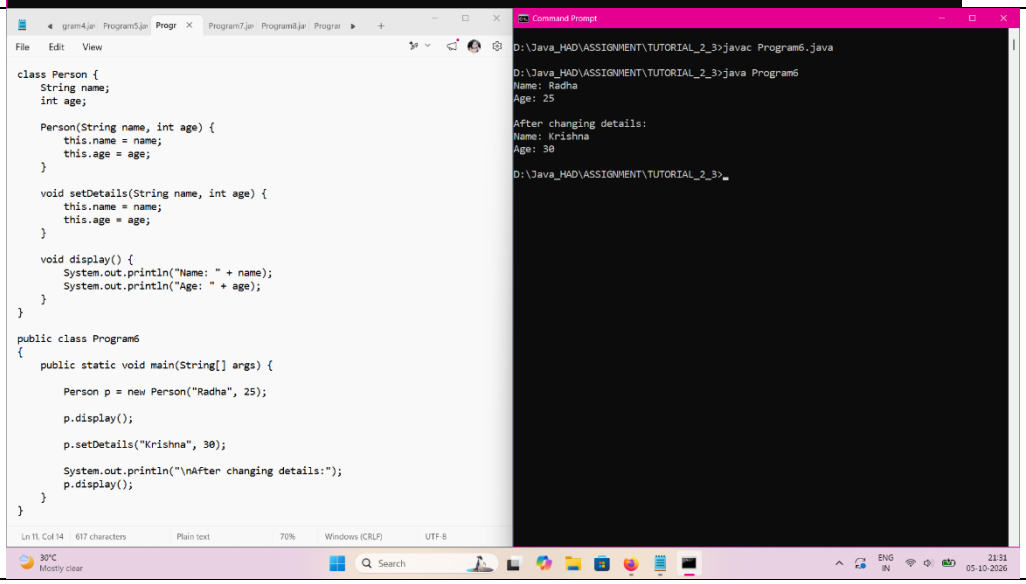
Output :

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program6.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program6
Name: Radha
Age: 25

After changing details:
Name: Krishna
Age: 30
```

Main Output :



Que 7 : Method Chaining (this Returning Current Object)

Code :

```
class Calculator {
  int result = 0;

  Calculator add(int number) {
    result += number;
```

```
    return this;
  }

  Calculator subtract(int number) {
    result -= number;
    return this;
  }

  Calculator multiply(int number) {
    result *= number;
    return this;
  }

  void display() {
    System.out.println("Result: " + result);
  }
}

public class Program7 {
  public static void main(String[] args) {

    Calculator calc = new Calculator();

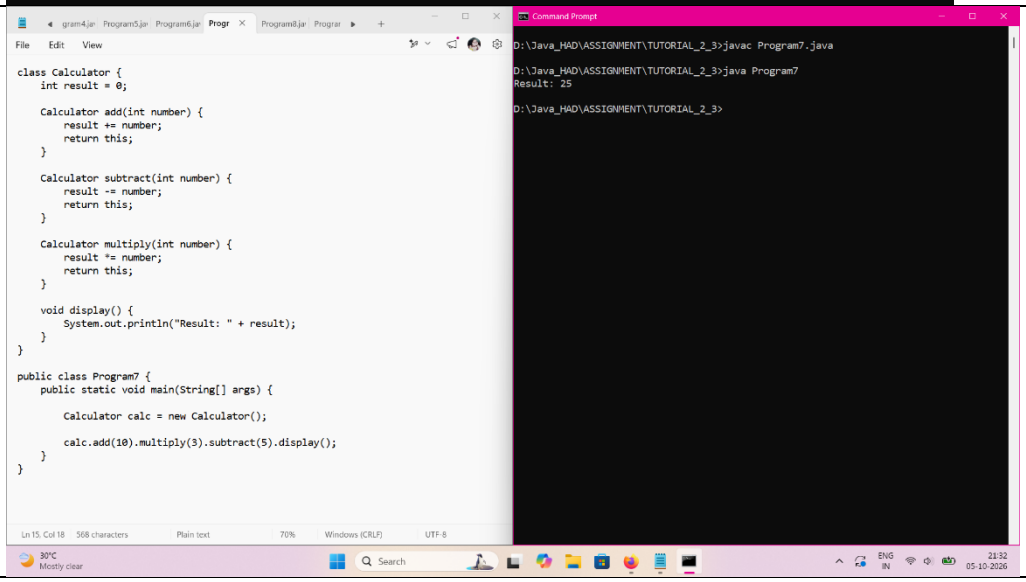
    calc.add(10).multiply(3).subtract(5).display();
  }
}
```

Output :

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program7.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program7
Result: 25
```

Main Output :



Que 8 : **Passing and Returning Objects**

Code :

```
class Complex {
  int real;
  int imaginary;

  Complex(int real, int imaginary) {
    this.real = real;
```

```
    this.imaginary = imaginary;
  }

  Complex add(Complex c) {
    return new Complex(
      real + c.real,
      imaginary + c.imaginary
    );
  }

  boolean isEqual(Complex c) {
    return real == c.real && imaginary == c.imaginary;
  }

  void display() {
    System.out.println(real + " + " + imaginary + "i");
  }
}

public class Program8 {

  static void change(Complex c) {
    c.real = 100;
  }

  public static void main(String[] args) {

    Complex c1 = new Complex(10, 20);
    Complex c2 = new Complex(5, 10);

    Complex c3 = c1.add(c2);

    System.out.print("Addition: ");
    c3.display();

    System.out.println("Are c1 and c2 equal? " + c1.isEqual(c2));

    change(c1);

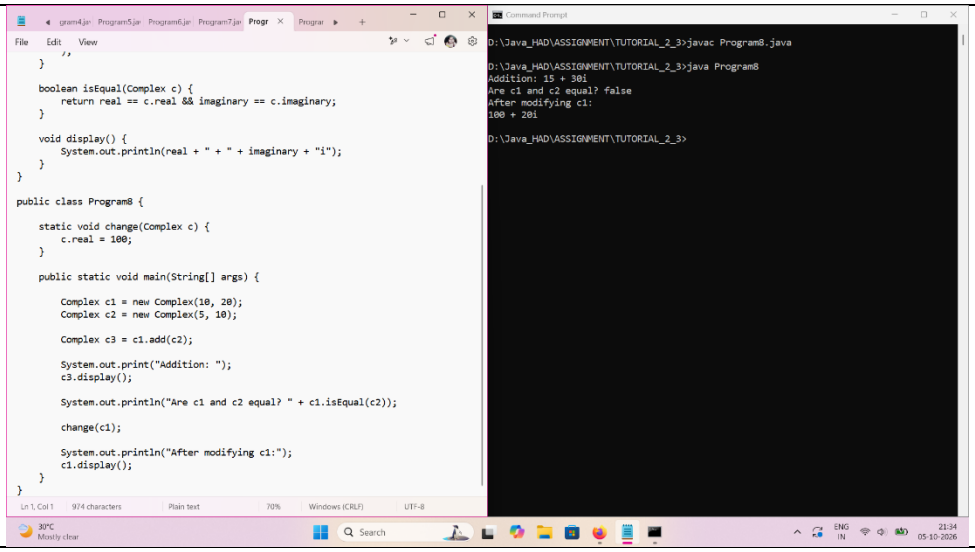
    System.out.println("After modifying c1:");
    c1.display();
  }
}
```

Output :

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program8.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program8
Addition: 15 + 30i
Are c1 and c2 equal? false
After modifying c1:
100 + 20i
```

Main Output :



```
    }

    boolean isEqual(Complex c) {
        return real == c.real && imaginary == c.imaginary;
    }

    void display() {
        System.out.println(real + " + " + imaginary + "i");
    }
}

public class Program8 {

    static void change(Complex c) {
        c.real = 100;
    }

    public static void main(String[] args) {

        Complex c1 = new Complex(10, 20);
        Complex c2 = new Complex(5, 10);

        Complex c3 = c1.add(c2);

        System.out.print("Addition: ");
        c3.display();

        System.out.println("Are c1 and c2 equal? " + c1.isEqual(c2));

        change(c1);

        System.out.println("After modifying c1:");
        c1.display();
    }
}
```

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program8.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program8
Addition: 15 + 30i
Are c1 and c2 equal? false
After modifying c1:
100 + 20i

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>
```

Que 9 : **Array of Objects and Sorting**

Code :

```
class Product {
  int id;
  String name;
  double price;

  Product(int id, String name, double price) {
    this.id = id;
    this.name = name;
    this.price = price;
  }

  void display() {
    System.out.println(id + "  " + name + "  " + price);
  }
}

public class Program9 {
  public static void main(String[] args) {

    Product[] products = {
      new Product(1, "Laptop", 50000),
      new Product(2, "Mobile", 20000),
      new Product(3, "Headphones", 3000),
      new Product(4, "Tablet", 25000),
      new Product(5, "Watch", 5000)
    };

    System.out.println("All Products:");
    for (Product p : products) {
      p.display();
    }

    Product costly = products[0];

    for (Product p : products) {
      if (p.price > costly.price) {
        costly = p;
      }
    }
```

```
        System.out.println("\nCostliest Product:");
        costly.display();

        // Bubble sort by price
        for (int i = 0; i < products.length - 1; i++) {
          for (int j = 0; j < products.length - i - 1; j++) {

            if (products[j].price > products[j + 1].price) {
              Product temp = products[j];
              products[j] = products[j + 1];
              products[j + 1] = temp;
            }
          }
        }

        System.out.println("\nProducts Sorted by Price:");

        for (Product p : products) {
          p.display();
        }
      }
    }
```

Output :

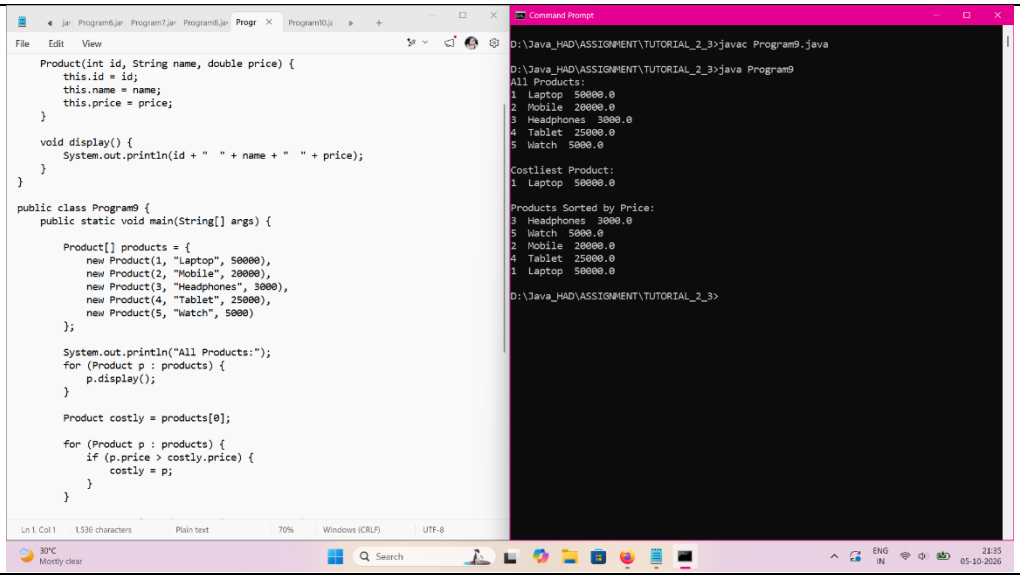
```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program9.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program9
All Products:
1  Laptop  50000.0
2  Mobile  20000.0
3  Headphones  3000.0
4  Tablet  25000.0
5  Watch  5000.0

Costliest Product:
1  Laptop  50000.0

Products Sorted by Price:
3  Headphones  3000.0
5  Watch  5000.0
2  Mobile  20000.0
4  Tablet  25000.0
1  Laptop  50000.0
```

Main Output :



```
    Product(int id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }

    void display() {
        System.out.println(id + "  " + name + "  " + price);
    }
}

public class Program9 {
    public static void main(String[] args) {

        Product[] products = {
            new Product(1, "Laptop", 50000),
            new Product(2, "Mobile", 20000),
            new Product(3, "Headphones", 3000),
            new Product(4, "Tablet", 25000),
            new Product(5, "Watch", 5000)
        };

        System.out.println("All Products:");
        for (Product p : products) {
            p.display();
        }

        Product costly = products[0];

        for (Product p : products) {
            if (p.price > costly.price) {
                costly = p;
            }
        }
```

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program9.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program9
All Products:
1  Laptop  50000.0
2  Mobile  20000.0
3  Headphones  3000.0
4  Tablet  25000.0
5  Watch  5000.0

Costliest Product:
1  Laptop  50000.0

Products Sorted by Price:
3  Headphones  3000.0
5  Watch  5000.0
2  Mobile  20000.0
4  Tablet  25000.0
1  Laptop  50000.0

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>
```

Que 10 : **Immutable Class and Singleton (Design Concepts)**

Code :

```
final class Address {
  private final String city;
  private final String state;

  Address(String city, String state) {
    this.city = city;
    this.state = state;
  }

  @Override
  public String toString() {
    return city + ", " + state;
  }
}

public class Program10 {
  public static void main(String[] args) {

    Address address = new Address("Ahmedabad", "Gujarat");

    System.out.println("Address: " + address);

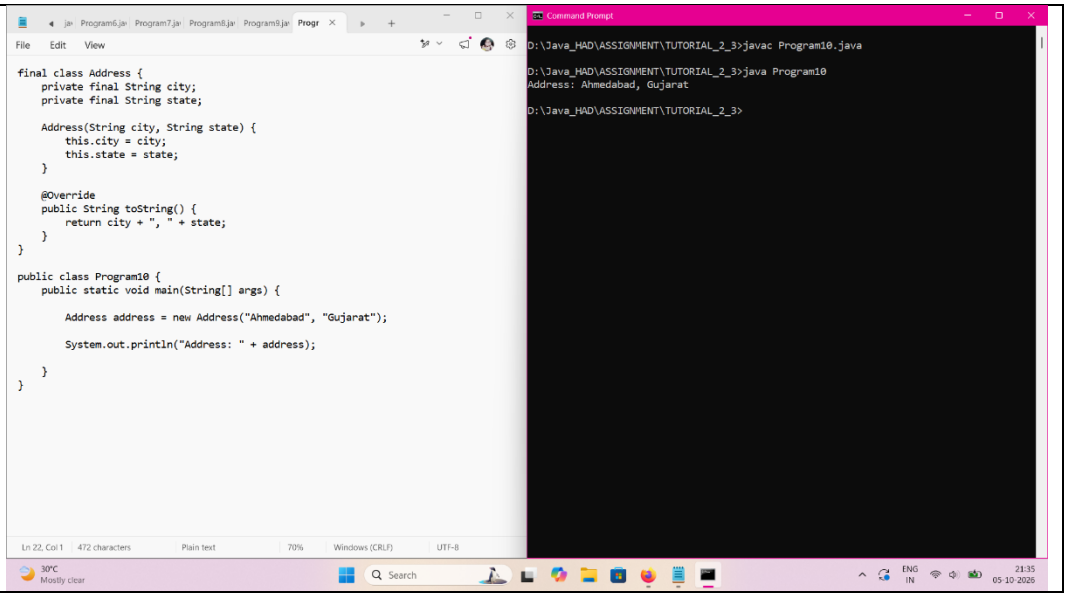
  }
}
```

Output :

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program10.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program10
Address: Ahmedabad, Gujarat
```

Main Output :



```
final class Address {
    private final String city;
    private final String state;

    Address(String city, String state) {
        this.city = city;
        this.state = state;
    }

    @Override
    public String toString() {
        return city + ", " + state;
    }
}

public class Program10 {
    public static void main(String[] args) {

        Address address = new Address("Ahmedabad", "Gujarat");

        System.out.println("Address: " + address);

    }
}
```

```
D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>javac Program10.java

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>java Program10
Address: Ahmedabad, Gujarat

D:\Java_HAD\ASSIGNMENT\TUTORIAL_2_3>
```